

Compute Operations

Operations Advisory Team

Office of CTO | EMEA Technology Engineering

May 2026

Agenda

1. Compute Concepts
2. Event-driven Automation
3. Provisioning Operations Scenarios
4. Change Operations Scenarios
5. Maintenance Operations Scenarios

INSTANCE TYPES

OCI Instance Types

There are 3 types of Instances in OCI:

Bare Metal (BM) Instance



dedicated physical server access for highest performance and strong isolation.

Virtual Machine (VM) Instance



a hypervisor on top of Bare Metal server provides Virtual Machine instances.

VMs are ideal for running applications that do not require the performance and resources (CPU, memory, network bandwidth, storage) of an entire physical machine.

Dedicated VM Host



A physical server reserved for a single customer, allowing to operate VM instances in isolation from other tenants.

Autoscaling & Instance pool is not supported in these instances.

INSTANCE COMPONENTS

OCI Instance components

Boot Volume (OS Disk).

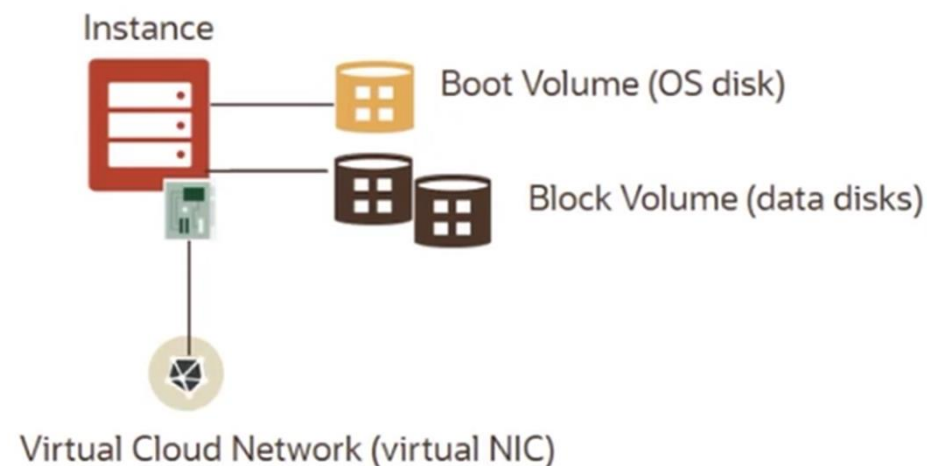
- A detachable boot volume device that contains the image used to boot a Compute instance.
- You can attach 1 per instance.

Block Volume (Data disks).

- A detachable block storage device that allows you to dynamically expand the storage capacity of an instance.
- You can attach 1-32 per instance. **Size:** from **50 GB** to **32 TB** in **1 GB** increments. Default size **1 TB**.
- **Attachment Types:** iSCSI or Paravirtualized.
- **Access types:** Read/write, Read/write shareable or Read-only

VNICs

- 1 primary private IPv4 address.
- Up to 64(32) optional secondary private IPv4(IPv6) addresses.
- An optional public ip for each private.



Block Volume Performance

Ultra High Performance

The Ultra High Performance level is recommended for workloads with the highest I/O requirements, requiring the best possible performance, such as large databases.

This option provides the best linear performance scale with 225 IOPS/GB up to a maximum of 300,000 IOPS per volume. Throughput also scales at the highest rate at 1,800 KB/s/GB up to a maximum of 2,680 MB/s per volume.

Balanced Performance (Default value)

The Balanced performance level provides a good balance between performance and cost savings for most workloads, including those that perform random I/O such as boot volumes.

This option provides linear performance scaling with 60 IOPS/GB up to 25,000 IOPS per volume. Throughput scales at 480 KB/s/GB up to a maximum of 480 MB/s per volume.

Higher Performance

The Higher Performance is recommended for workloads

with high I/O requirements that don't require the performance of the Ultra High Performance level.

This option provides a linear performance scale of 75 IOPS/GB up to a maximum of 50,000 IOPS per volume. Throughput scales at the rate 600 KB/s/GB up to a maximum of 680 MB/s per volume.

Lower performance

The Lower Cost elastic performance option is recommended for throughput intensive workloads with large sequential I/O, such as streaming, log processing, and data warehouses.

This option gives you linear scaling 2 IOPS/GB up to a maximum of 3000 IOPS per volume. Throughput scales at 240 KB/s/GB up to the maximum of 480 MB/s per volume.

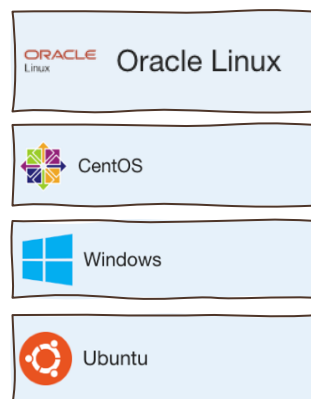
IMAGES

COMPUTE IMAGES



An image is a template of a virtual hard drive, determines the operating system and other software for an instance.

Platform Images



Platform images

Oracle Linux, Ubuntu, CentOS, Windows,

Pre-built Oracle images



Software price per OCPU

\$0.4646/hr

Your actual costs depends on various factors. ⓘ

Oracle Images and complete solutions enabled for OCI

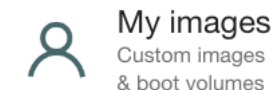
Like OEM, Oracle Database, WLS suite, etc

Third-party images



Trusted **third-party images** published by Oracle partners from the Partner Image catalog. RedHat, Rocky Linux, Alma Linux

Custom Images



☒ Custom images ☐ Boot volumes ☐ Image OCID

- Custom images: Platform + software
- Boot volumes
- RED HAT is now supported.
 - ☐ KVM Guest Images, which are downloaded from the RHEL access download page.
 - ☐ QCOW2-formatted images, which are downloaded from the Red Hat Image Builder tool.

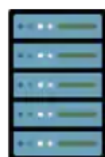


SHAPES

OCI Instance Shapes

Oracle Cloud Infrastructure offers a variety of [shapes](#) that are designed to meet a range of compute and application requirements:

Flexible & Standard Shapes



Flexible shapes let users customize the number of OCPUs and the amount of memory when provisioning or resizing a VM.

Standard shapes are designed for general purpose workloads and provide a balance of cores, memory and network resources.

Flexible and Standard Shapes are available with Intel, AMD and ARM processors

DenseIO Shapes



Designed for **large databases, big data workloads, and applications that require high-performance local storage.**

DenseIO shapes include locally-attached NVMe-based SSDs.

GPU Shapes



Designed for **hardware-accelerated workloads.**

GPU shapes include Intel or AMD CPUs and NVIDIA graphics processors.

High performance computing (HPC) and Optimized Shapes



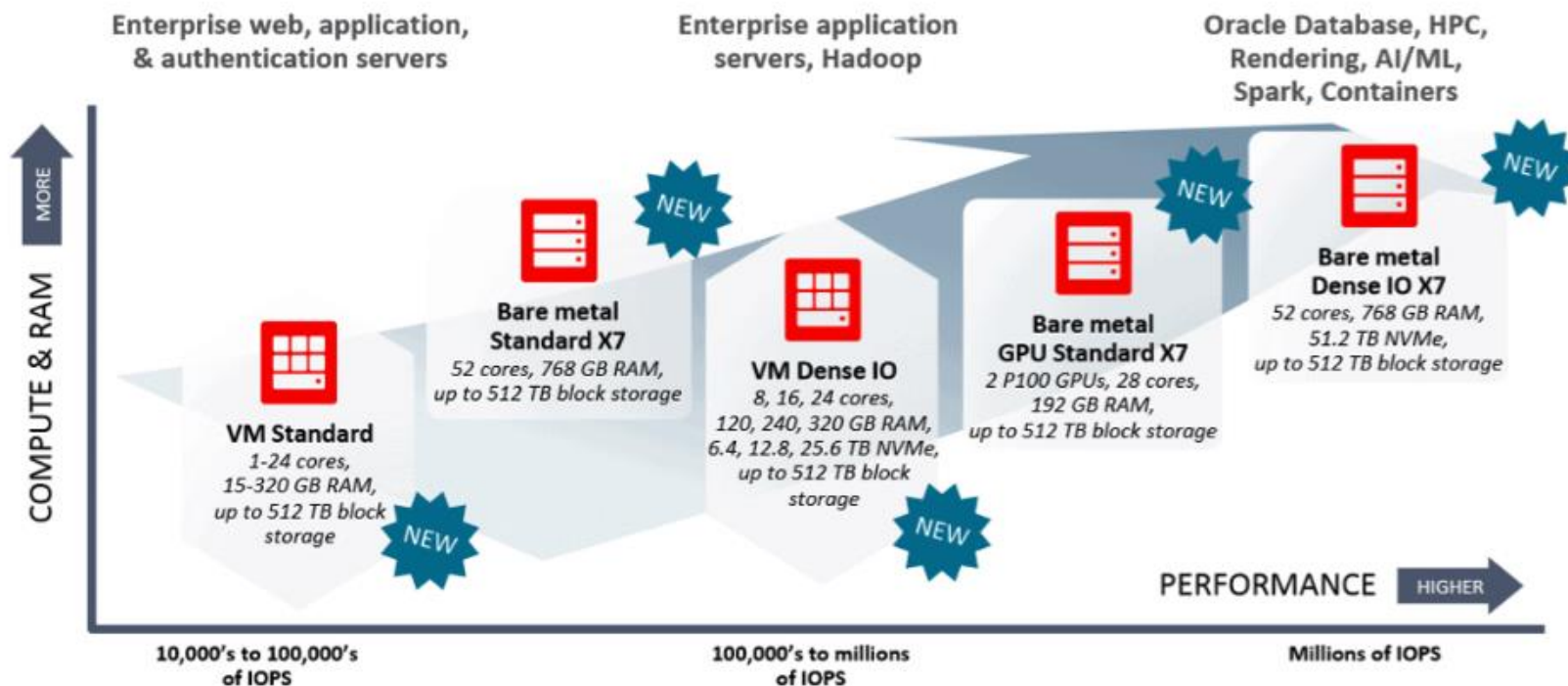
Designed for **high-performance computing workloads that require high frequency processor cores.**

Bare metal HPC and optimized shapes support cluster networking.

FLEXIBLE SHAPES

A flexible shape is a shape that lets you **customize the number of OCPUs and the amount of memory** when launching or resizing your VM.

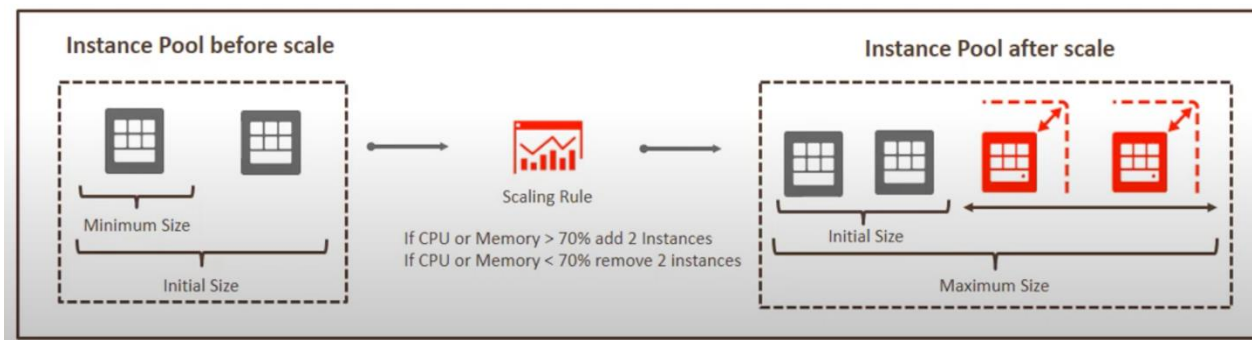
This flexibility lets you build VMs that match your workload, enabling you to optimize performance and minimize cost.



INSTANCE POOL

INSTANCE POOL

Use instance pools to create and manage multiple compute instances as a group. (within the same region).



Oracle Cloud autoscaling automatically adjusts the number of Compute instances in an instance pool based on performance metrics such as CPU utilization, memory utilization. This helps provide consistent performance for end-users during periods of high demand, and helps reduce costs during periods of low demand.

You can apply the following types of autoscaling to an instance pool:

- **Metric-based autoscaling**: An autoscaling action is triggered when a performance metric meets or exceeds a threshold.
- **Schedule-based autoscaling**: Autoscaling events take place at the specific times that you schedule.

Autoscaling is supported for virtual machine (VM) and bare metal instance pools that use standard, dense I/O, and GPU shapes.

SPECIALIZED WORKLOADS

OCI specialized compute workloads

Oracle Cloud Infrastructure offers features that let you customize your instances for specialized workloads and security requirements.

Burstable instances are virtual machine (VM) instances that provides a baseline level of CPU performance with the ability to burst to a higher level to support occasional spikes in usage. (Flex shapes)

Use cases:

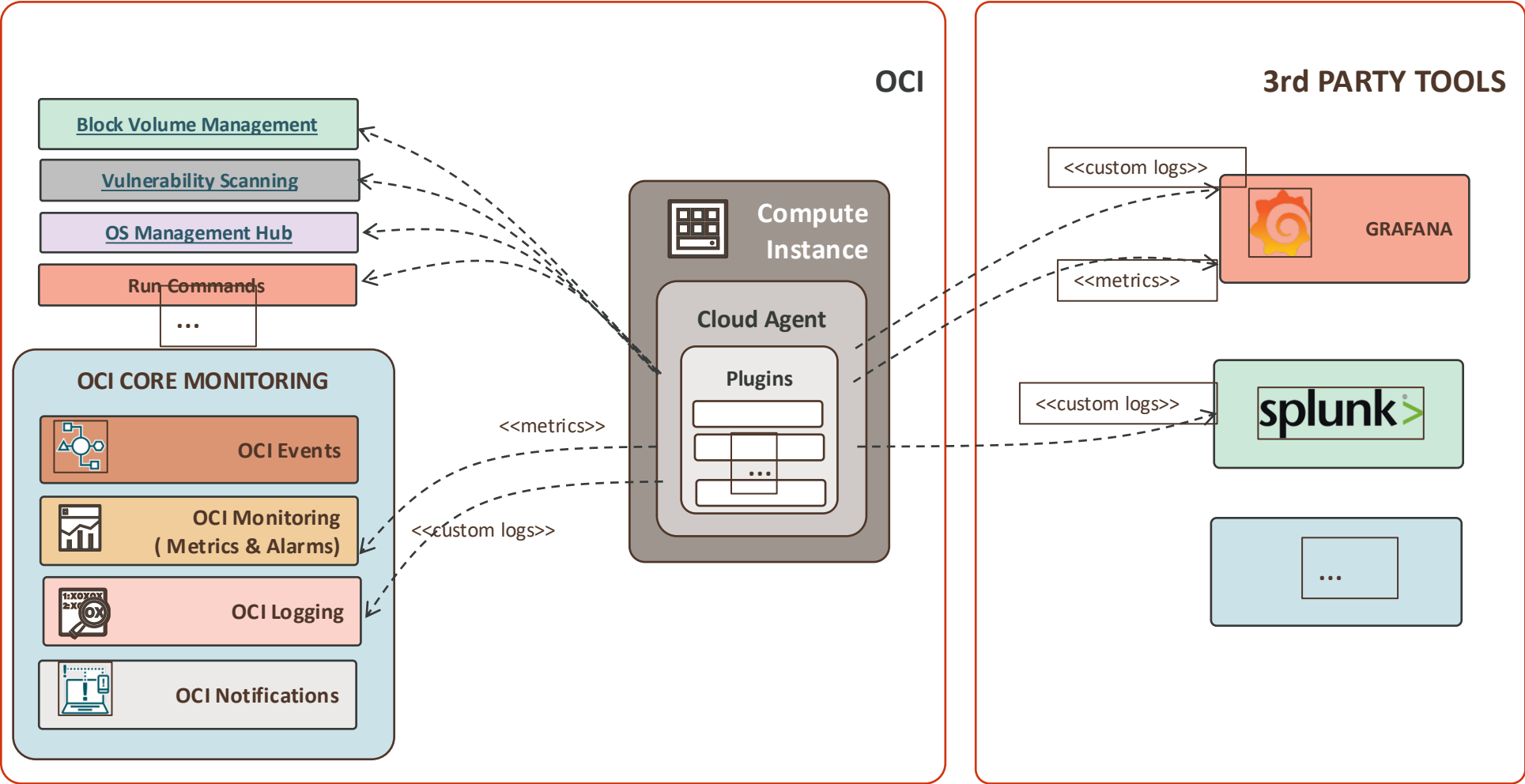
- Microservices
- Development and test environments
- Continuous integration and continuous delivery (CI/CD) tools
- Monitoring systems
- Static websites

Shielded instances harden the firmware security on bare metal hosts and virtual machines (VMs) to defend against malicious boot level software.

Shielded instances use the combination of Secure Boot, Measured Boot, and the Trusted Platform Module to harden the firmware security on your instances.

CLOUD AGENT AND PLUG-INS

Cloud Agents & Plug-ins



OCI Monitoring | OCI Compute Monitoring



- **Oracle Cloud Agent** is a lightweight process that manages plugins running on compute instances. Oracle Cloud Agent is supported on current **platform images and on custom images** that are based on current platform images. Oracle Cloud Agent is installed by default on current platform images.
- **Plugins** collect performance metrics, install OS updates, and perform other instance management tasks. To use plugins on an instance, the Oracle Cloud Agent software must be installed on the instance, the plugins must be enabled, and the plugins must be running. You might need to perform additional configuration tasks before you can use certain plugins.

OCI Compute Monitoring: plug-ins

Plug-in	Enable by Default	Description
OS Management Hub Agent		Manages updates and patches for the operating system environment on the instance.
Compute Instance Run Command		Runs scripts within the instance to remotely configure, manage, and troubleshoot the instance.
Custom Logs Monitoring		Ingests custom logs into the Logging service.
Compute Instance Monitoring		Emits metrics about the instance's health, capacity, and performance. These metrics are consumed by the Monitoring service.
Management Agent		Collects data from resources such as OSs, applications, and infrastructure resources for OCI services that are integrated with Management Agent. Data can include observability, log, configuration, capacity, and health data.
Oracle Autonomous Linux		Manages autonomous updates and collects data associated with events, including logs and stack traces, for instances managed by the Autonomous Linux service



OCI Compute Monitoring: plug-ins



Plug-in

Plug-in	Enable by Default	Description
Vulnerability Scanning	X	Scans the instance for security vulnerabilities.
Oracle Java Management Service	X	Monitors and performs Java Development Kit (JDK) lifecycle management for Java deployments on instances managed by the Java Management service.
Block Volume Management	X	Configures block volume sessions within the instance.
Bastion	X	Allows secure shell (SSH) connections to an instance without public IP addresses using the Bastion service.

OCI Cloud agent plug-ins | Console view 1of2



Plug-in

Oracle Cloud Agent

[Oracle Cloud Agent](#) is a lightweight process that manages plugins running on the instance. Plugins collect performance metrics, install OS updates, and perform other instance management tasks.

Q Search and Filter

Search

Stop plugins

Plugin name ↑↓	Enable plugin ↑↓	Status ↑↓	Last updated ↑↓	Message ↑↓	Plugin description ↑↓	
WebLogic Management Service	Disabled	—	May 2, 2025, 04:35 UTC	-	Monitors WebLogic Servers running on compute instances, manages WebLogic Se	...
Vulnerability Scanning	Disabled	—	May 2, 2025, 04:35 UTC	-	Scans the instance for security vulnerabilities.	...
Oracle Java Management Service	Disabled	—	May 2, 2025, 04:35 UTC	-	Monitors and performs Java Development Kit (JDK) lifecycle management for Java	...
Oracle Autonomous Linux	Enabled	Stopped	May 2, 2025, 04:35 UTC	-	Monitors events and delivers updates to instances managed by the Oracle Autono	...
OS Management Service Agent	Enabled	Not supported	May 2, 2025, 04:35 UTC	-	Manages updates and patches for the operating system environment on the instan	...
OS Management Hub Agent	Disabled	—	May 2, 2025, 04:35 UTC	-	Manages and monitors updates and patches for the operating system environmen	...
Management Agent	Disabled	—	May 2, 2025, 04:35 UTC	-	Collects data from resources such as OSs, applications, and infrastructure resource	...
Custom Logs Monitoring	Enabled	Running	May 2, 2025, 04:35 UTC	-	Ingests custom logs into the Logging service.	...
Compute RDMA GPU Monitoring	Disabled	—	May 2, 2025, 04:35 UTC	-	The plugin handles metrics and faults for RDMA and GPU.	...
Compute Instance Run Command	Enabled	Running	May 2, 2025, 04:35 UTC	-	Runs scripts within the instance to remotely configure, manage, and troubleshoot t	...

◀

▶

Page 1 of 2 (1 - 10 of 16 total items)

Items per page10



OCI Cloud agent plug-ins | Console view 2of2



Plug-in

Oracle Cloud Agent

[Oracle Cloud Agent](#) is a lightweight process that manages plugins running on the instance. Plugins collect performance metrics, install OS updates, and perform other instance management tasks.

Q Search and Filter

Search

Stop plugins

Plugin name ↕	Enable plugin ↕	Status ↕	Last updated ↕	Message ↕	Plugin description ↕	
Compute Instance Monitoring	Enabled	Running	May 2, 2025, 04:35 UTC	-	Emits metrics about the instance's health, capacity, and performance.	...
Compute HPC RDMA Auto-Configuration	Disabled	—	May 2, 2025, 04:35 UTC	-	The plugin handles configuration for RDMA network and firmware settings for	...
Compute HPC RDMA Authentication	Disabled	—	May 2, 2025, 04:35 UTC	-	The plugin handles RDMA authentication	...
Cloud Guard Workload Protection	Disabled	—	May 2, 2025, 04:35 UTC	-	Manages the lifecycle of Workload Protection resources.	...
Block Volume Management	Disabled	—	May 2, 2025, 04:35 UTC	-	Configures block volume sessions within the instance.	...
Bastion	Disabled	—	May 2, 2025, 04:35 UTC	-	Allows secure shell (SSH) connections to an instance without public IP address	...

◀

▶

Page 2 of 2 (11 - 16 of 16 total items)

Items per page 10 ▼



VULNERABILITY SCANNING SERVICE

OCI Compute Monitoring: Vulnerability Scanning

By default



Oracle Cloud Infrastructure [Vulnerability Scanning Service](#) helps improve your security posture by routinely checking hosts and container images for potential vulnerabilities. Watch a [video introduction](#) to the service.

The Scanning service can identify several types of security issues:

Ports

Ports that are unintentionally left open might be a potential attack vector to your cloud resources, or enable hackers to exploit other vulnerabilities

OS Packages

OS packages that require updates and patches to address vulnerabilities.

OS conf

OS configurations that hackers might exploit.

Benchmarks

Industry-standard benchmarks published by the [Center for Internet Security](#) (CIS).

The Scanning service checks hosts for compliance with the section 5 (*Access, Authentication, and Authorization*) benchmarks defined for [Distribution Independent Linux](#).

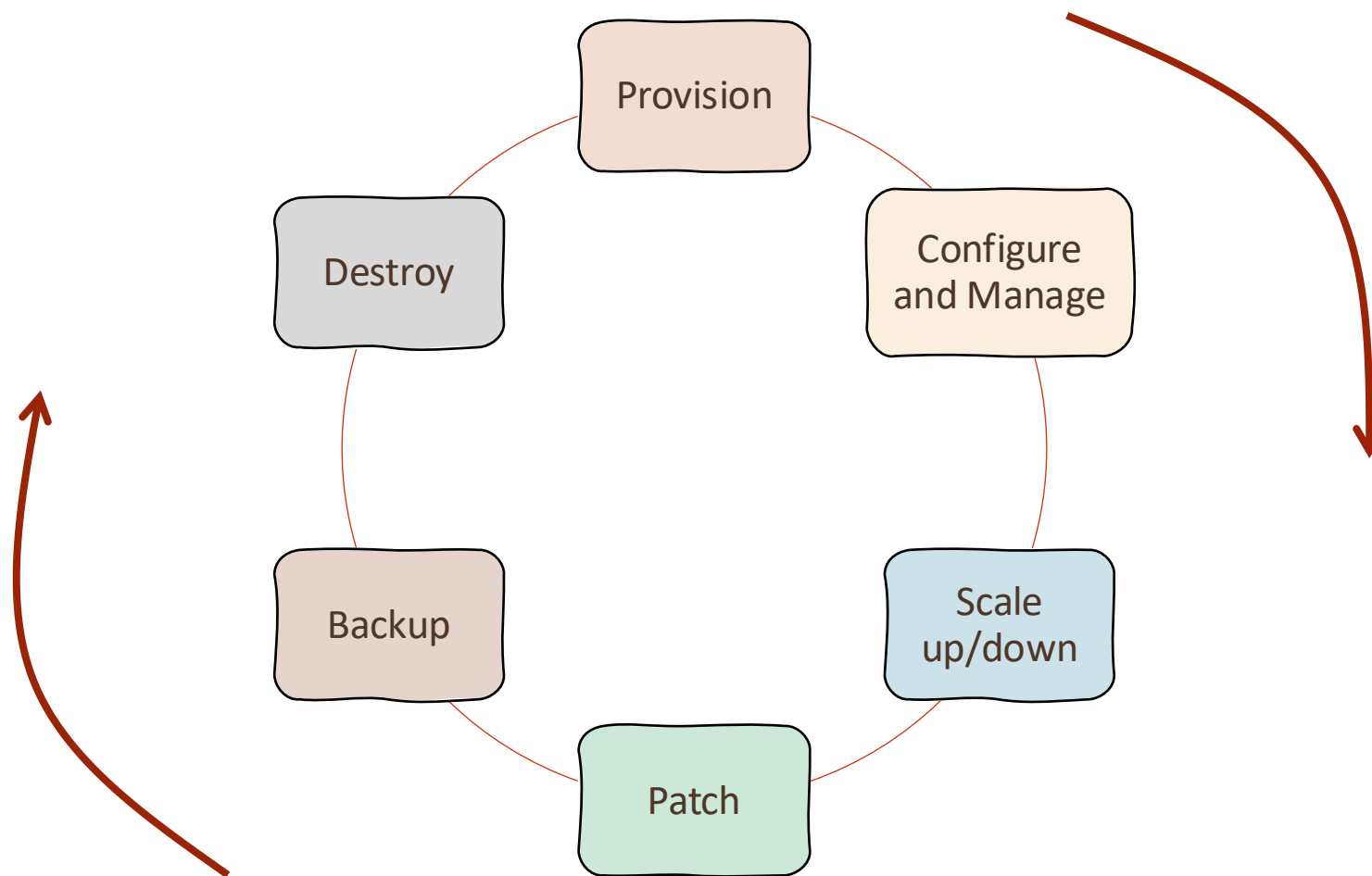
App

Vulnerabilities in third-party applications such as log4j and spring4shell.

[Cloud Guard](#) is an Oracle Cloud Infrastructure service that provides a central dashboard to monitor all of your cloud resources for security weaknesses in configuration, metrics, and logs. When it detects a problem, it can suggest, assist, or take corrective actions, based on your Cloud Guard configuration.

COMPUTE LIFECYCLE OVERVIEW

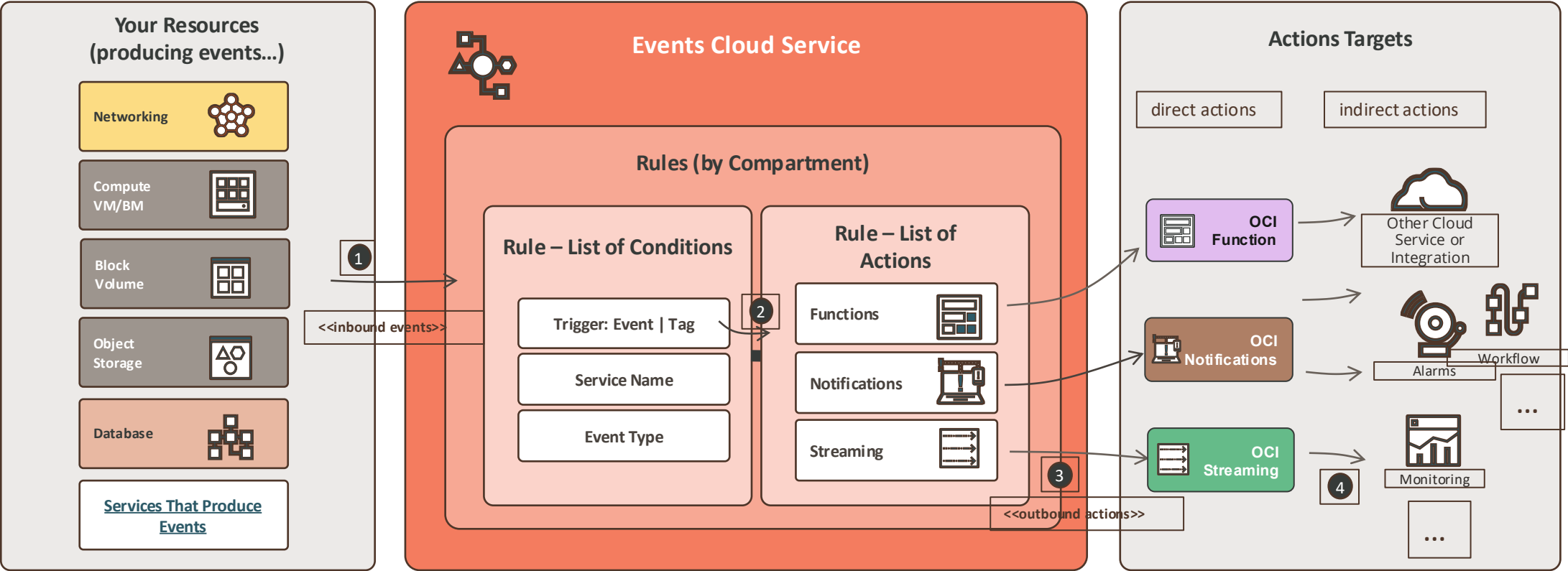
OCI Compute Monitoring: Identify compute operations





EVENT DRIVEN AUTOMATION

OCI Events Cloud Service | Event-Driven Operations with CNCF*



OCI Compute Monitoring: Configure and Manage computes

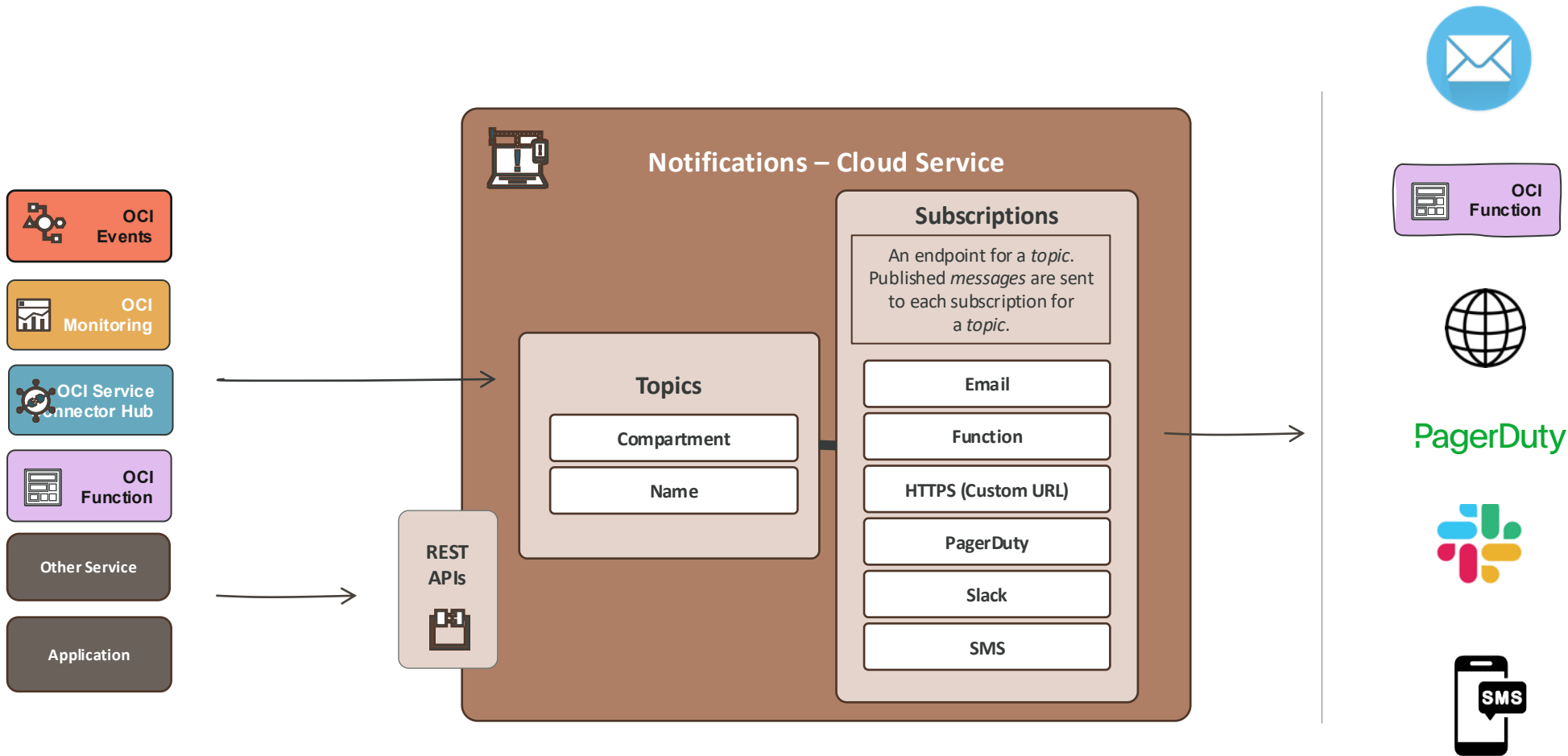
Events-Driven Automation

You can create automation based on state changes for your Oracle Cloud Infrastructure resources by using event types, rules, and actions.

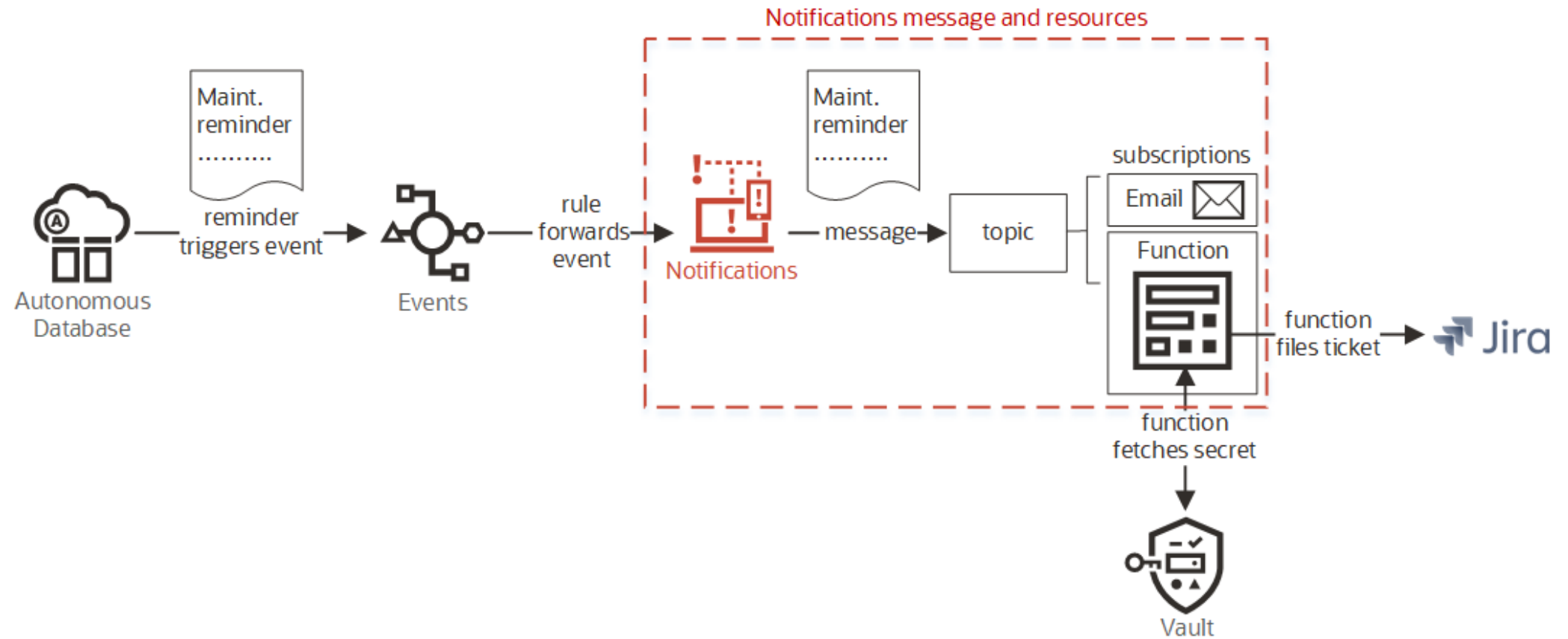
The following Compute resources emit events:

- [Autoscaling Event Types](#)
- [Capacity Reservation Event Types](#)
- [Cluster Network Event Types](#)
- [Console History Event Types](#)
- [Image Event Types](#)
- [Instance Event Types](#)
- [Instance Configuration Event Types](#)
- [Instance Console Connection Event Types](#)
- [Instance Maintenance Event Types](#)
- [Instance Pool Event Types](#)

OCI Notifications Cloud Service | Basic Integrations



OCI Events & Notification | Example | Opening Tickets



PROVISIONING & DESTROYING OPERATION SCENARIOS

Provisioning Operations Scenarios

- In this section we'll talk about the following Provisioning Operations Scenarios:
 - Provision Instances.
 - Configure Instances during provisioning.
 - Destroy/Terminate Instances.

PROVISIONING COMPUTE INSTANCES

Provisioning Instances | Console



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Instance Configurat

Instance Pools

Instances

An [instance](#) is a compute host. Choose between virtual machines (VMs) and bare metal instances. The image that you use to launch an instance de

Search and Filter

Applied filters

Compartment Ralf_lange

Create instance

Actions

	Name	State	Public IP	Private IP	Shape	OCPU count	Memory (GB)
☐	emcc	Running	89.168.108.122	10.0.0.187	VM.Standard.E4.Flex	2	32
☐	unmanaged	Stopped	130.61.235.197	10.10.117.30	VM.Standard.A2.Flex	0.5	6
☐	ol9x86	Stopped	158.180.23.254	10.10.31.180	VM.Optimized3.Flex	1	14
☐	ol8x86	Stopped	130.61.179.255	10.10.67.198	VM.Optimized3.Flex	1	14
☐	ol9amd	Stopped	130.61.112.214	10.10.6.127	VM.Standard.E4.Flex	1	16



Provisioning Instances | Console



Create compute instance

Create an instance to deploy and run applications, or save as a reusable Terraform stack for creating an instance with Resource Manager.

Name

instance-20230208-1630

Create in compartment

PJUAREZ_ES

oci4cca (root)/CCA_Basic_Compartment/PJUAREZ_ES

Placement

The availability domain helps determine which shapes are available.

Availability domain

AD 1

NoEK:EU-FRANKFURT-1-AD-1

✓

AD 2

NoEK:EU-FRANKFURT-1-AD-2

AD 3

NoEK:EU-FRANKFURT-1-AD-3

Show advanced options

Security

Shielded instance: Disabled
Confidential computing: Disabled

Image and shape

A [shape](#) is a template that determines the number of CPUs, amount of memory, and other resources allocated to an instance. The image is the operating system that runs on top of the shape.

image

ORACLE
Linux

Oracle Linux 8

Image build: 2023.01.31-2

🛡️🔒🔑

Change image

Shape

AMD

VM.Standard.E4.Flex

Virtual machine, 1 core OCPU, 16 GB memory, 1 Gbps network bandwidth

🛡️🔒

Change shape

Provisioning Instances | Console

Select an image

 Oracle Linux ✓

 Ubuntu

 CentOS

 Windows

 SUSE

 AlmaLinux

 Rocky Linux

 Marketplace

 My images
Custom images
& boot volumes

Compartment

PJUAREZ_ES

oci4cca (root)/CCA_Basic_Compartment/PJUAREZ_ES

Search...			
Image name	Publisher	Price	Security
☐ Oracle Autonomous Linux 7.9	Oracle	Messages.instances.launch.priceTypes.FREE()	  ▼
☐ Oracle Linux 6.10	Oracle	Messages.instances.launch.priceTypes.FREE()	
☐ Oracle Linux 7.9	Oracle	Messages.instances.launch.priceTypes.FREE()	   ▼
☒ Oracle Linux 8	Oracle	Messages.instances.launch.priceTypes.FREE()	   ▼
☐ Oracle Linux 9	Oracle	Messages.instances.launch.priceTypes.FREE()	  ▼
☐ Oracle Linux Cloud Developer 8	Oracle	Messages.instances.launch.priceTypes.FREE()	 ▼
1 Selected			
Showing 6 Items < 1 of 1 >			

Provisioning Instances: Console

Networking

[Collapse](#)

[Networking](#) is how your instance connects to the internet and other resources in the Console. To make sure you can [connect to your instance](#), assign a public IP address to the instance.

Primary network

☒ Select existing virtual cloud network
 ☐ Create new virtual cloud network
 ☐ Enter subnet OCID

Virtual cloud network in **PJUAREZ_ES** [\(Change Compartment\)](#)

vcn_test

Subnet

An IP address from a public subnet and an [internet gateway](#) on the VCN are required to make this instance accessible from the internet.

☒ Select existing subnet
 ☐ Create new public subnet

Subnet in **PJUAREZ_ES** [\(Change Compartment\)](#)

EMDBSubnet (regional)

Public IPv4 address

☒ Assign a public IPv4 address
 ☐ Do not assign a public IPv4 address

 If you're not sure whether you need a public IP address, you can always assign one later.

[Show advanced options](#)

Add SSH keys

Generate an [SSH key pair](#) to connect to the instance using a Secure Shell (SSH) connection, or upload a public key that you already have.

☒ Generate a key pair for me
 ☐ Upload public key files (.pub)
 ☐ Paste public keys
 ☐ No SSH keys

 Download the private key so that you can connect to the instance using SSH. It will not be shown again.

[Save private key](#)
[Save public key](#)

Boot volume

A [boot volume](#) is a detachable device that contains the image used to boot the compute instance.

☐ Specify a custom boot volume size
[Volume performance](#) varies with volume size. Default boot volume size: 46.6 GB. When you specify a custom boot volume size, service limits apply.

☒ Use in-transit encryption
[Encrypts data](#) in transit between the instance, the boot volume, and the block volumes.

☐ Encrypt this volume with a key that you manage
 By default, Oracle manages the keys that encrypt this volume, but you can choose a key from a vault that you have access to if you want greater control over the key's lifecycle and how it's used. [How do I manage my own encryption keys?](#)



Provisioning Instances | OCI CLI – REST API

[oci compute instance launch](#) [OPTIONS]

--availability-domain [text]
--compartment-id, -c [text]
--shape [text]
--subnet-id [text]

[Core Services API](#)

Provisioning Instances: Terraform

To deploy a compute you need to use a resource with type oci_core_instance

When you deploy the instance using terraform you can enable/disable plug-ins during the provisioning phase.

```
resource "oci_core_instance" "deploy_instance" {

  #Required
  availability_domain = var.domain
  compartment_id     = var.compartment_id
  shape              = var.shape_vm

  agent_config {
    are_all_plugins_disabled = "false"
    is_management_disabled  = "false"
    is_monitoring_disabled  = "false"

    plugins_config {
      desired_state = var.plug_in_vul
      name          = "Vulnerability Scanning"
    }
    plugins_config {
      desired_state = var.plug_in_java
      name          = "Oracle Java Management Service"
    }
  }
  .....

  plugins_config {
    desired_state = var.plug_in_mon
    name          = "Compute Instance Monitoring"
  }
}
```

CONFIGURING COMPUTES

OCI Compute Monitoring: Configure and Manage computes

During deployment:

- **Custom Image.**

Deploy a pre-build image with all the required configuration.

- [Cloud-init. \(linux \) / Cloudbase-init \(windows\).](#)

Is a software package that automates the initialization of cloud instances during system boot.

- [Terraform provisioners.](#)

You can use provisioners to model specific actions on the local machine or on a remote machine in order to prepare servers or other infrastructure objects for service.

DESTROY / TERMINATE INSTANCES

Destroy Computes | Console



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Cloud Native Database

Instances

An [instance](#) is a compute host. Choose between virtual machines (VMs) and bare metal instances. The image that you use to launch an instance de

Search and Filter

Applied filters Compartment **Ralf_lange**

Create instance

Actions

Reboot

Start

Stop

Terminate

	Name	Private IP	Shape	OCPU count	Memory (GB)		
☒	emcc	8.122	10.0.0.187	VM.Standard.E4.Flex	2	32	
☐	unmanage	5.197	10.10.117.30	VM.Standard.A2.Flex	0.5	6	
☐	ol9x86	Stopped	136.160.23.254	10.10.31.180	VM.Optimized3.Flex	1	14
☐	ol8x86	Stopped	130.61.179.255	10.10.67.198	VM.Optimized3.Flex	1	14
☐	ol9amd	Stopped	130.61.112.214	10.10.6.127	VM.Standard.E4.Flex	1	16



Destroy Computes | OCI CLI | REST API | TERRAFORM

```
oci compute instance terminate [OPTIONS]  
--instance-id [text]
```

[Core Services API](#)

[Terraform Destroy](#)

CHANGE OPERATION SCENARIOS

Change Operations Scenarios

- In this list you can see some examples of change Operations Scenarios:
 - [Stopping and starting and Instance](#)
 - [Changing performance of a Block Volume](#)
 - [Adding users to an Instance](#)
 - [Getting Instance Metadata & Updating Instance Metadata](#)
 - [Move instance to other compartment](#)
 - [Change the shape](#)
 - [Rename an Instance](#)
 - [Attaching a Volume](#)
 - [Add multiple VNICs](#)
 - ...

COMPUTE INSTANCE RUN COMMAND

OCI Compute Monitoring: Run Command

By default



Plug-in

- You can remotely **configure, manage, and troubleshoot compute** instances by running scripts within the instance using the run command feature.
- You can run commands on an instance even when the instance does not have SSH access or open inbound ports.

OCI CONSOLE

REST API

OCI Console:
Compute>
Instances >
Instance Details >
Run Command

OCI Compute Monitoring: Run Command

By default



Plug-in

Limitations and Considerations

- On Linux instances, the script runs in a Bash shell by default. To run the script with a different program, use `#!/<path_to_program>` as the first line of the script.
- On Windows instances, the script runs in a batch shell by default. To run the script with PowerShell, use `#ps1` as the first line of the script.
- The maximum size for a script file that you upload directly to an instance in plain text is 4 KB. To provide a larger file, save the file in an Object Storage location.
- The output of a script when returned as plain text is limited to the last 1 KB. To save a larger response, save the output to an Object Storage location.
- Two scripts can run at a time by default. (Customized)
- A maximum of five scripts can be in flight at a time.
- Scripts that prompt for information are not supported
- The maximum time that a script can run is one day.
- To monitor the resources that scripts consume, such as CPU utilization, use [metrics](#).
- For troubleshooting review log `/var/log/oracle-cloud agent/plugins/runcommand/runcommand.log`

OCI Compute Monitoring: Run Command

By default



Plug-in

Prerequisites:

1. Cloud Agent installed in the compute instance (default for platform images)
2. Plug-in Run Command enabled (default)
3. Set up OCI IAM Policies :

- Group:

<Your Group>

- Dynamic group:

<dynamic_group_name> . Any {instance.compartment.id = *<Compartment>*}

- IAM policies

Allow group *<Your Group>* to manage instance-agent-command-family in compartment *<Compartment>*

Allow dynamic-group *<dynamic_group_name>* to use instance-agent-command-execution-family in compartment *<compartment>*

Allow dynamic-group *<dynamic_group_name>* to read objects in compartment *<compartment>*
(where all {target.bucket.name = '*<bucket_with_script_file>*'})

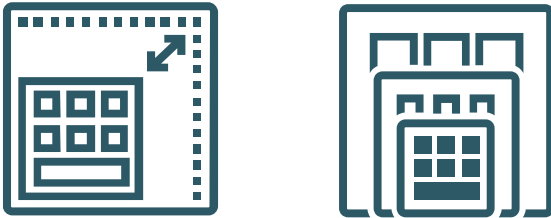
Allow dynamic-group *<dynamic_group_name>* to manage objects in compartment *<compartment>*
(where all {target.bucket.name = '*<bucket_for_command_output>*'})

COMPUTE SCALING

OCI Compute Instances Scaling Options

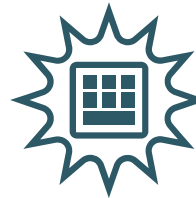
There are basically 3 options to scale quickly compute VM instances in OCI:

Change shape



- This is a **vertical, cold scaling** option
- Changing the shape implies to **stop** the VM and use its boot volume to **start** it in a different Host
- It is possible also to use **Flexible shapes**, which don't change the host but add additional cores (and/or memory), from the same host

Burstable Instances



- This is a **vertical, hot scaling** option
- Workloads runs at a fraction of CPUs most of the time, burst up to the full CPUs for a limited amount of time when needed
- No guarantee that an instance will be able to burst when needed

Instance pools



- This is a **horizontal, hot scaling** option
- When a CPU (or memory) threshold is surpassed, a new instance is provisioned. Optionally, it can be added to the Load Balancer backend pool
- When the CPU (or memory) gets back to normal values, the unneeded instance is removed from the pool

Changing Shape Automation Example

- We can automate the compute VM shape change by using different methods, like OCI CLI/SDK or any other REST API based script
- In OCI we don't have any native scheduler (yet), but we could run that script in any administration VM and schedule it by using, for example, cron
- Example: Using OCI CLI to change to a flexible or standard shape amount of OCPUs

```
$ oci compute instance update --instance-id <OCID of the compute instance> [--shape-config  
file://path/to/file/scaleUp.json | --shape <shape type>]
```

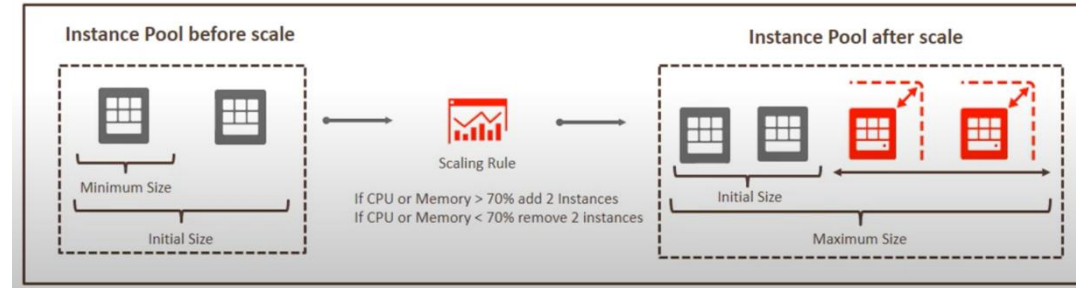
- The JSON file looks like:

```
{  
  "baselineOcpuUtilization": "BASELINE_1_1",  
  "memoryInGBs": 16.0,  
  "ocpus": 1.0  
}
```

- A shape type example could be "VM.Standard2.1"
- When you run the command, a work request will be generated and the instance will stop and start again using the new shape

Instance Pools

- Use instance pools to create and manage multiple compute instances as a group (within the same region)



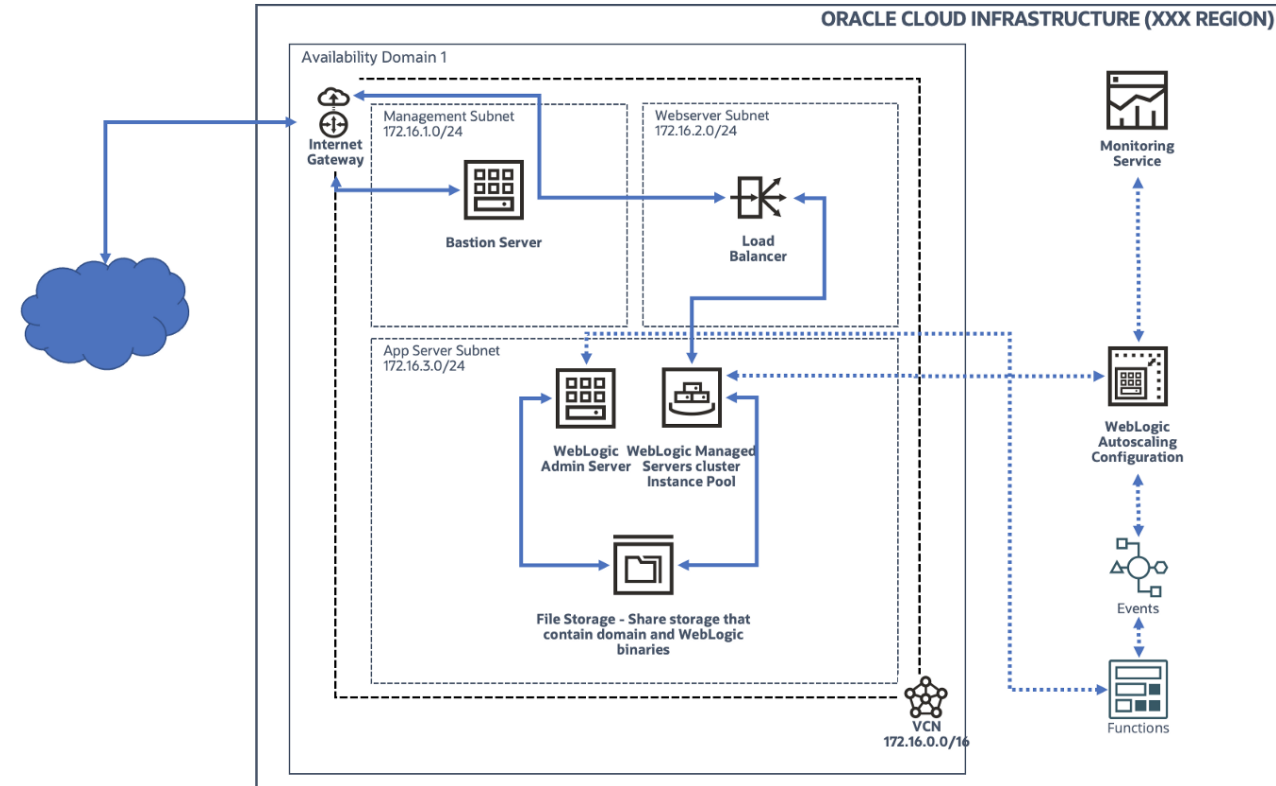
- Autoscaling lets us automatically adjust the number of Compute instances in an instance pool based **on performance metrics** such as CPU or memory utilization or **on time schedule-based**. This helps us provide consistent performance for end-users during periods of high demand, and helps reduce your costs during periods of low demand

You can apply the following types of autoscaling to an instance pool:

- **Metric-based autoscaling**: An autoscaling action is triggered when a performance metric meets or exceeds a threshold
- **Schedule-based autoscaling**: Autoscaling events occur at the specific times you schedule
- Supported for virtual machine (VM) and bare metal instance pools that use standard, dense I/O, and GPU shapes
- This autoscaling option is convenient for workloads with loose coupling, like web servers. WLS servers can use it but it needs from a more complex autoscaling solution based on OCI functions to include/remove managed servers in the cluster

Instance Pools

- What can we do in a WLS environment?. [Example](#):
- **Scaling up:**
 - We can have an instance pool with a CPU based metric autoscaling policy or time-based scheduling
 - If CPU $\geq$ X% or time schedule is met, a new instance will be created based in an instance configuration (custom boot image)
 - New instance is registered in the load balancer
 - New instance mounts a File Storage mount point with the WLS config
 - Linux systemd executes custom WLST script, after the instance has started successfully, to create a new managed server and adds it to the cluster
- **Scaling down:**
 - If user traffic decreases or time-schedule for scale down is met, OCI monitoring rises scale-in event
 - OCI autoscaling terminates one of the instances
 - Instance terminate event triggers an OCI Function
 - OCI Function will execute a custom script inside the WLS admin server to stop and remove the managed server from the WLS cluster



MAINTENANCE OPERATION SCENARIOS

Maintenance Operations Scenarios

- In this list you can see some examples of maintenance Operations Scenarios:
 - Patch OS Compute. OS Management Hub Service
 - Backup & Restore

PATCH OS COMPUTE

OCI Compute Monitoring: Patch computes

You can patch:

- 1) Manually (YUM/DNF update)
- 2) Automate using OS Management Hub Service (Console | OCI CLI | API | SDK)

OS Management service allows you to **manage , monitor updates and patches** for the operating system environments on your Oracle Cloud instances, including instances managed by the OS Management Oracle Autonomous Linux service.

OCI Compute Monitoring | OS Management Hub Service



Plug-in

- Prerequisites:
 1. Cloud Agent installed in the compute instance (default for platform images)
 2. Plug-in OS Management Hub enabled (not enabled by default)
 3. Set up OCI IAM Policies for the OS Management Hub service :
 - **Dynamic Group:**
Any {instance.compartment.id = '<compartment>', instance.compartment.id = '...'}
 - **Let OS Management Hub manage instances in the dynamic group:**
Allow dynamic-group <osmh-instances> to {OSMH_MANAGED_INSTANCE_ACCESS} in tenancy where request.principal.id = target.managed-instance.id
 - **Let users manage OS Management Hub resources in the tenancy:**
Allow group <osmh-admins> to manage osmh-family in tenancy
 - **Let users manage OS Management Hub resources in a compartment:**
Allow group <osmh-admins> to read osmh-profiles in tenancy where target.profile.compartment.id = '<tenancy-ocid>'
Allow group <osmh-admins> to read osmh-software-sources in tenancy where target.softwareSource.compartment.id = '<tenancy-ocid>'
Allow group <osmh-admins> to manage osmh-family in compartment <compartment>

OCI Compute Monitoring: OS Management Hub Service



Plug-in

Cloud

Search resources, services, document

Germany Central (Frankfurt)

Search

Home

Compute

Storage

Networking

Oracle Database

Databases

Analytics & AI

Developer Services

Identity & Security

Observability & Management

Hybrid

Migration & Disaster Recovery

Lifecycle Management

Actions and Controls

Administration

Monitoring

Service Metrics

Metrics Explorer

Alarm Status

Agent Configurations

Alarm Definitions

Health Checks

Logging Analytics

Home

Log Explorer

Solutions

Dashboards

Administration

Events Service

Events

Jobs

Reports

Profiles

OS Management Hub

Overview

Software sources

Groups

Lifecycle environments

Instances

Events

Jobs

Reports

Profiles

Entitlements

Management stations

Monitored resources

Resource Analytics

OCI Compute Monitoring: OS Management Hub Service



Or directly from a specific instance:

← Instances

ol8x86

Running

Instance details

Details

Networking

Storage

Security

Management

OS Management

Monitoring

Work requests

Tags

Overview

Use the OS Management Hub service to manage updates and patches on compute instances.

Available updates

Security 6 Bug 2 Enhancement 0 Other 0

Operating system

Oracle Linux 8.10

Kernel

5.15.0-306.177.4.1.el8uek.x86_64

Effective kernel

5.15.0-306.177.4.1.el8uek.x86_64

Scheduled jobs

0

Details

[View OS Management Hub details](#)

Instances

An [instance](#) is a compute host. Choose between virtual machines (VMs) and other software.

Search and Filter

Applied filters: Compartment [Raif_lange](#)

Create instance

Actions

☐	Name ↑↓	State ↑↓	Public IP ↑↓	Private IP
☐	emcc	Running	89.168.108.122	10.0.1.1
☐	unmanaged	Running	130.61.235.197	10.0.1.2
☐	ol9x86	Running	158.180.23.254	10.0.1.3
☐	ol8x86	Running	130.61.179.255	10.0.1.4



OCI Compute Monitoring: OS Management Hub Automation



Plug-in

List all the computes available in a compartment:

```
oci os-management-hub managed-instance list --all --compartment-id $OCID
```

Update all packages on managed instances in compartment which are not part of a group.

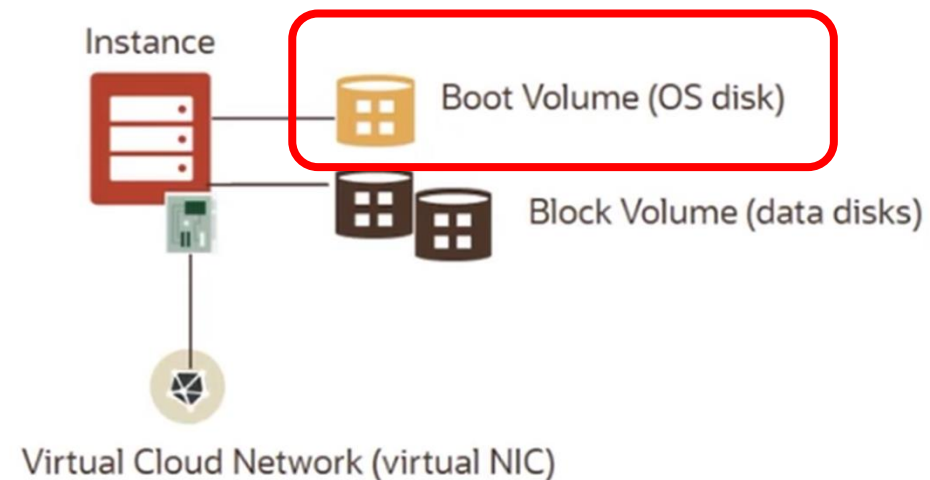
```
oci os-management-hub update-all-packages-in-compartment --compartment-id $OCID
```

BACKUP & RESTORE INSTANCES

OCI Compute Operations: Boot Volume

Boot Volume Operations (Console | API | OCI CLI)

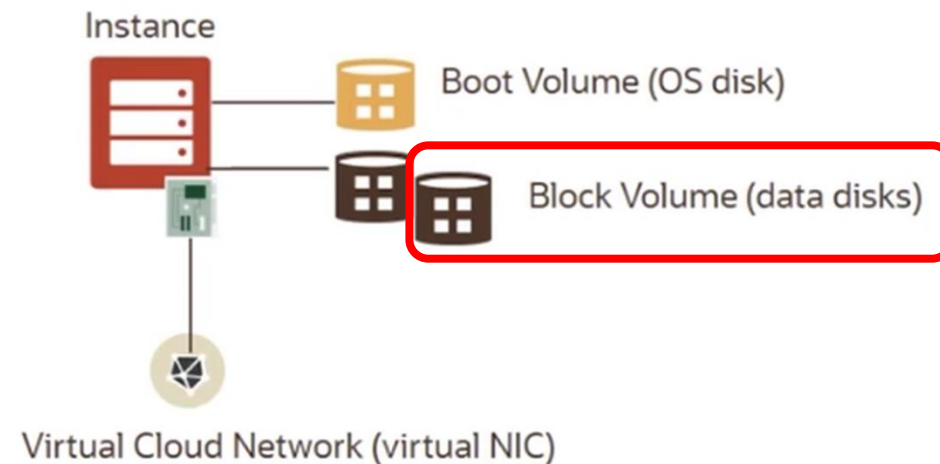
- [Backing Up a Boot Volume](#)
- [Restoring a Boot Volume](#)
- [Copying a Boot Volume Backup Between Regions](#)
- [Cloning a Boot Volume](#)
- [Detaching a Boot Volume](#)
- [Deleting a Boot Volume](#)
- [Replicating a Boot Volume](#)



OCI Compute Operations: Block Volume

Block Volume Operations (Console | API | OCI CLI)

- [Backing Up a Block Volume](#)
- [Restoring a Block Volume](#)
- [Copying a Block Volume Backup Between Regions](#)
- [Cloning a Block Volume](#)
- [Replicating a Block Volume](#)
- [Detaching a Block Volume](#)
- [Moving Block Volumes between Compartments](#)
- [Deleting a Block Volume](#)



OCI Compute Operations: Boot&Block Volume Backups Concepts

- A Backup of a Boot/Block Volume is a point-in-time snapshot of the data.
- Volume backups can be made while the volume is detached or attached.
- Create Manual or Policy-Based Backups.
- Full + incremental Backups.
- A Boot Volume can be created by restoring it from a Boot Volume Backup
- Copy Backups cross regions for Disaster Recovery or migration purposes.
- Volume Groups allow Backups of different volumes to be consistent at the same point-in-time
- [Overview of Boot Volume Backups](#)
- [Overview of Block Volume Backups](#)

Block/Boot Backup vs Clon

	Volume Backup	Volume Clone
Description	Creates a point-in-time backup of data on a volume. You can restore multiple new volumes from the backup later in the future.	Creates a single point-in-time copy of a volume without having to go through the backup and restore process.
Use case	Retain a backup of the data in a volume, so that you can duplicate an environment later or preserve the data for future use. Meet compliance and regulatory requirements, because the data in a backup remains unchanged over time. Support business continuity requirements. Reduce the risk of outages or data mutation over time.	Rapidly duplicate an existing environment. For example, you can use a clone to test configuration changes without impacting your production environment.
Speed	Slower (minutes or hours)	Faster (seconds)
Cost	Lower cost	Higher cost
Storage location	Object Storage	Block Volume
Retention policy	Policy-based backups expire, manual backups do not expire	No expiration
Volume groups	Supported. You can back up a volume group.	Supported. You can clone a volume group.

STORAGE REPLICATION

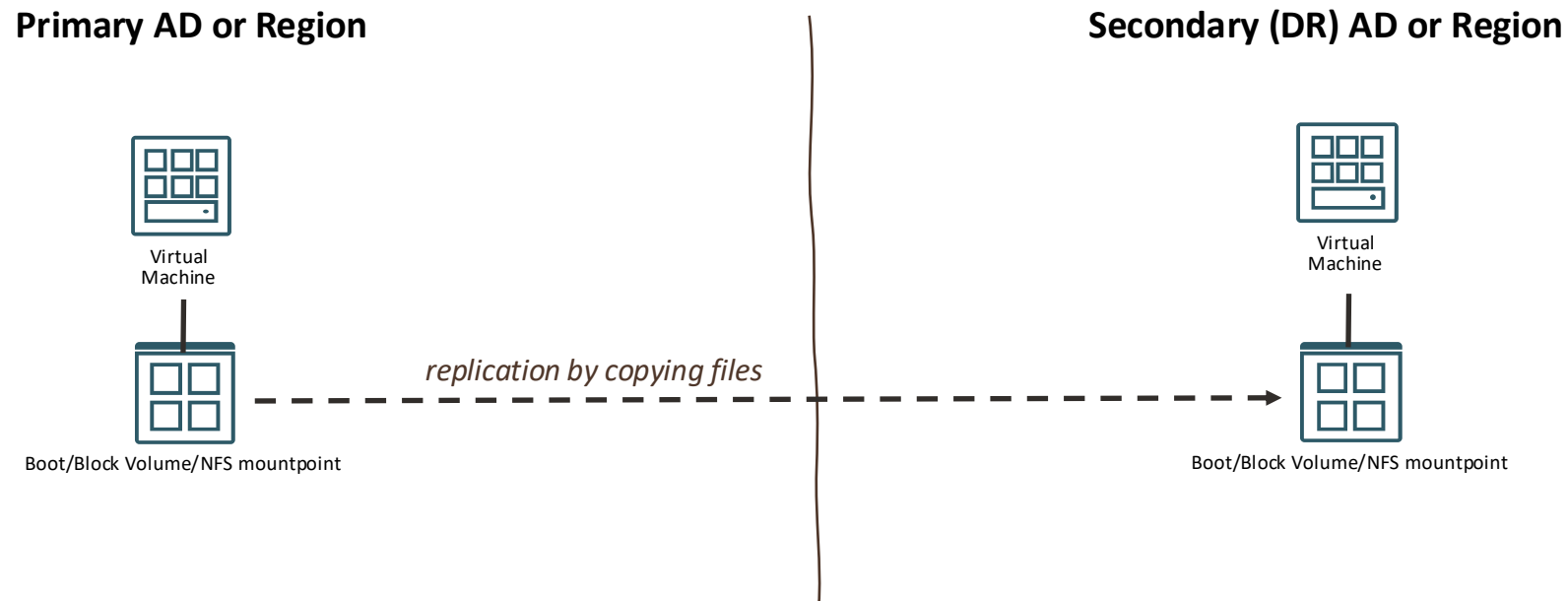
Storage Replication Options

Let's consider the following options:

1. Copy files periodically with tools (scp, rsync)
2. Boot volume replication
3. Block volume replication
4. File Storage replication

Copy the information with tools (rsync, scp)

- You can sync files between 2 systems inside the OS with tooling like rsync or scp (or other [parallel tools](#))
- This is also an asynchronous copy method
- You would need password-less authentication using public/private keys
- A scheduler is needed in order to copy periodically the information (like cron or other)
- You can sync full copies or just the delta information
- Works within between Availability Domains or across regions as long as a network connection between the instances is established



Block volume replication: Features

Boot Volume (OS disk)

- A detachable boot volume device that contains the image used to boot a Compute instance
- You can attach 1 per instance
- Local to the availability domain. Cannot be moved between ADs or Regions

Block Volume (Data disks)

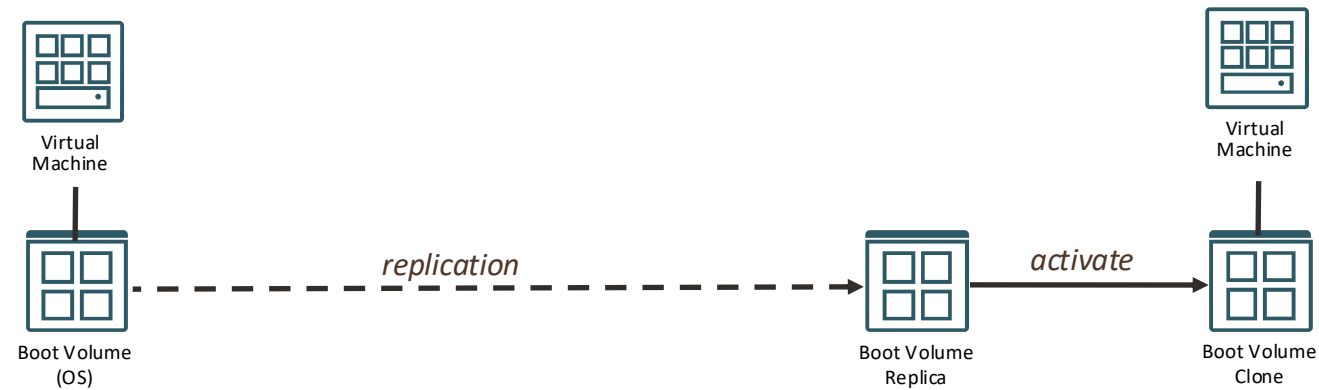
- A detachable block storage device that allows you to dynamically expand the storage capacity of an instance
- You can attach 1-32 per instance. **Size:** from **50 GB** to **32 TB** in **1 GB** increments. Default size **1 TB**
- **Attachment Types:** iSCSI or Paravirtualized
- **Access types:** Read/write, Read/write shareable or Read-only
- Local to the availability domain. Cannot be moved between ADs or Regions
- Volume backups can be restored to another region or AD
- To being able to share between different hosts, a cluster file system is needed to avoid data corruption (like OCFS2)
- Block volumes can be grouped in Block Volumes Groups for an easier management of replicas, backups, etc.

→ Both, boot and block volumes can be replicated **asynchronously** between ADs in the same region or between regions. This does not require volume backups.

Boot volume replication | How it works

Primary AD or Region

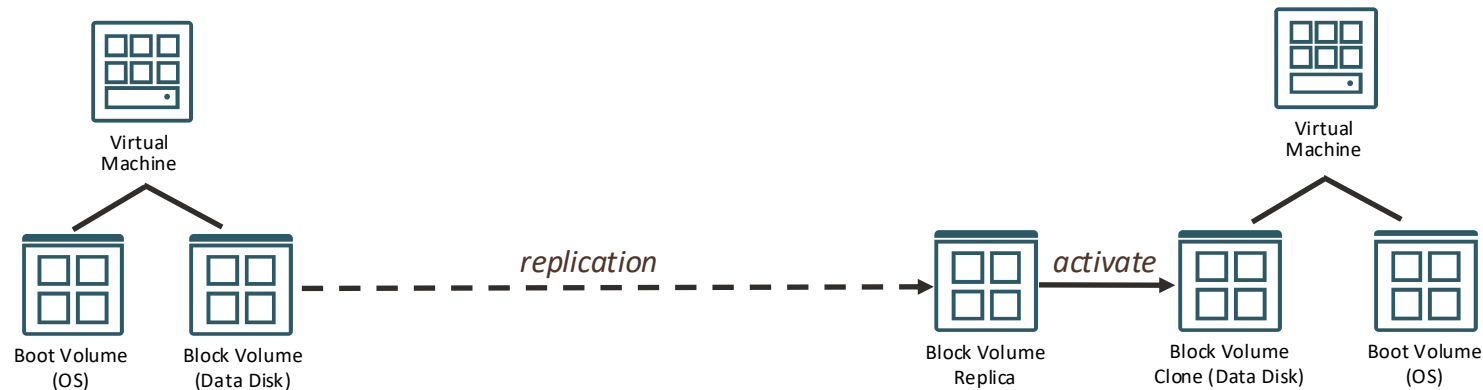
Secondary (DR) AD or Region



Block volume replication: How it works

Primary AD or Region

Secondary (DR) AD or Region



Block volume replication: Guidelines

- When you enable replication, an initial sync of data is performed from source to target, which depending on the volume size and amount of the information, can take up to hours to finish
- After the initial sync, the replication process is continuous
- Typical RPO target is less than 30 mins for cross-region replication but can vary depending on the change rate of data in the source volume
- Replication is not supported in volumes encrypted with customer-managed keys
- A replication-enabled volume cannot be resized. If you need to resize it, disable replication, resize and re-enable replication again
- Replication does not impact on the source volumes
- If you're using different block volumes, a block volume group replication must be used
- When you need to use the replicated block volume, you need to activate it, which will create a clone of the volume in the target AD/region. Then you can attach to the VM you want to use

Boot volume replication: How to setup

- 1 Menu → Storage → Boot Volumes → Set your compartment and select your boot volume
- 2 Edit the volume → Cross Region Replication → ON
- 3 Select same or different region, target Availability Domain and name for the Boot Volume Replica
- 4 Save Changes
- 5 Check Status and sync status on Boot Volume Replicas

Block volume replication: How to setup

- 1 Menu → Storage → Block Volumes → Set your compartment and select your boot volume
- 2 Edit the volume → Cross Region Replication → ON
- 3 Select same or different region, target Availability Domain and name for the Block Volume Replica
- 4 Save Changes
- 5 Check Status and sync status on Block Volume Replicas

NOTE: If you have a Block Volume which belongs to a Block Volume Group it is better to enable the replication for the whole group in Volume Groups

Block volume replication: How to activate

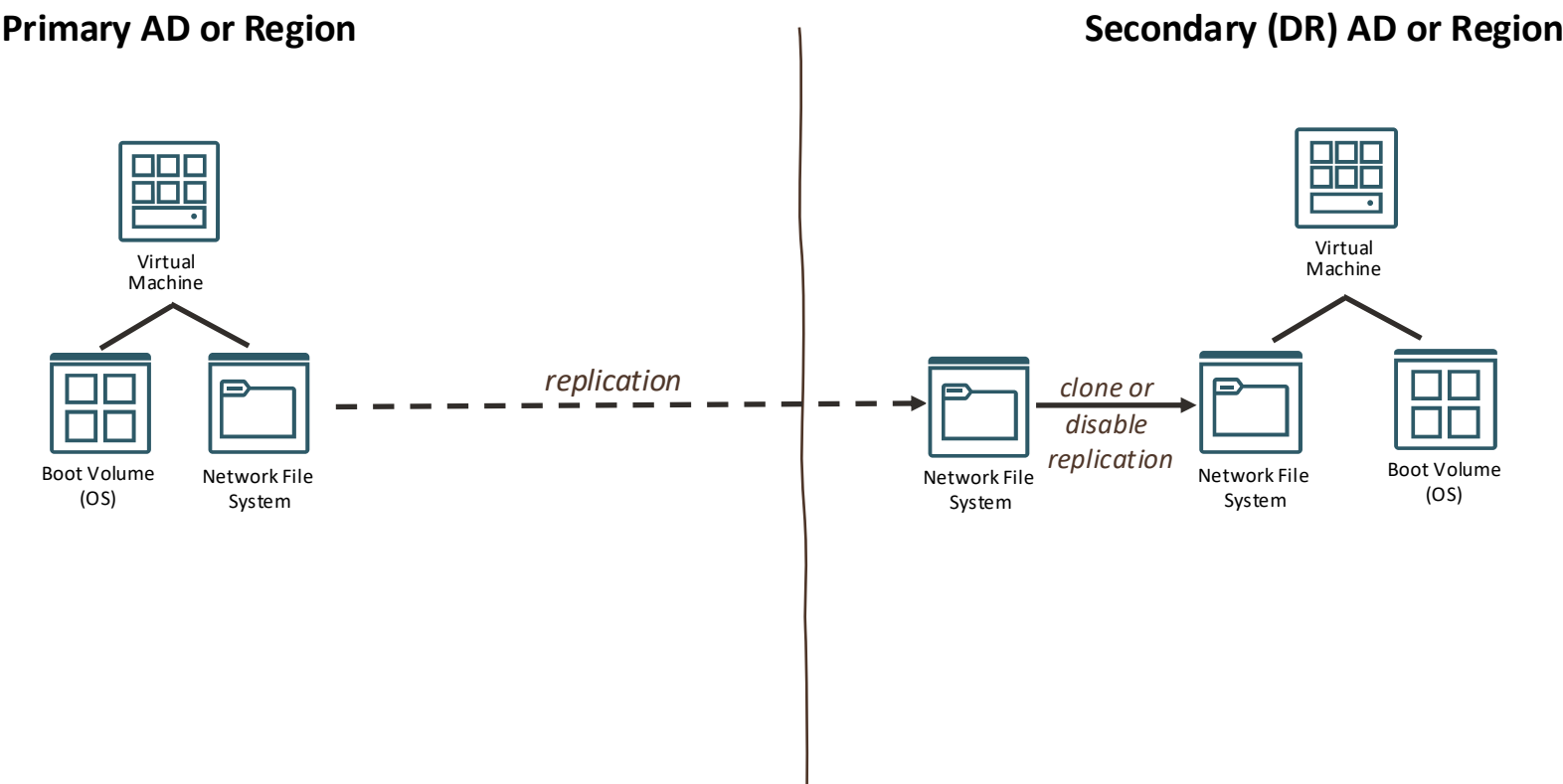
- 1 Menu → Storage → Block Volumes → Set your compartment and select your Block (or Boot) Volume Replicas
- 2 In your Block Volume Replica press the 3 dots symbol on the right and select **Activate** and confirm
- 3 Select a name for the clone, the destination compartment, and customize the volume details (size and perf)
- 4 Press Activate Replica
- 5 Select the new block volume clone in the Block Volumes resources page
- 6 To attach to a compute instance, select the Attached Instances → Attach to Instance
- 7 Select the attachment type as iSCSI and the needed access type (usually R/W) and the compute instance
- 8 (Optional) Attach the iSCSI Block Volume to the compute instance and run the discovery/attach OS commands
- 9 Mount the device on your filesystem

NOTE: It is recommended that you select the “Use Oracle Cloud Agent to automatically connect to iSCSI-attached volumes”. You’d need to enable Cloud Agent plug-in “Block Volume Management” in the compute instance. It will save to you some manual steps to discover the iSCSI volume in the compute instance

File Storage replication: Features

- Useful when different compute VMs need to read/write information from a storage repository, like in WLS clustered applications, thanks to its locking mechanism
- Supports NFSv3 with Network Lock Manager functionality
- Mount targets (NFS endpoints), lives in a subnet, AD specific
 - You can create (by default) 2 mount targets per account per AD
- Almost unlimited storage (cost per usage)
- We can configure file system **asynchronous replication** between ADs or cross-region
- We can customize the **replication interval**, this is the frequency that the replication operation is performed
- The replication is performed by taking a snapshot of the filesystem and transferring and applying the snapshot to the target, un-exported file system
- To use the replicated file system, we need to stop replication and then export it or create a clone file system from the latest snapshot
- A different mount target in the target AD/region is needed
- RPO is the last replication snapshot transmitted to the target
- Only some of the nearest regions are selected as [available target regions](#) for replication

File Storage replication: How it works



Storage Replication Options comparison

Feature	OS tools replication	Boot Volume Replication	Block Volume Replication	File Storage Replication
Sync method	Asynchronous	Asynchronous	Asynchronous	Asynchronous
Sync level	File	Block	Block	File
Access	Within OS	iSCSI	iSCSI	NFS
Needed infrastructure	Primary & Standby running	Just the primary running. Standby instantiated on switchover/failover	Primary and optionally standby running	Primary and optionally standby running
Automation	Needs external script to orchestrate	REST APIs support + OS script (to mount)	REST APIs support + OS script (to mount)	REST APIs support + OS script (to mount)
FSDR Support	As external script step	Native	Native	As external script step

THANKS

